

“I valued the free response problem sets so highly and couldn’t stress them enough for a comprehensive review leading up to the AP test. Since it is a year long course, these problem sets refreshed me on techniques and strategies that we learned in the first two months of the class and made me feel more prepared for what I would encounter on the AP test.” A 2019 Nerd

FR1. 1998 AB 4 (Calculator)

Let $y = f(x)$ be a function with $f(1) = -4$ such that for all points (x, y) on the graph of f the slope is given by $\frac{3x^2 - 1}{2y}$.

- Find the slope of the graph of f at the point where $x = 1$.
- Write an equation for the line tangent to the graph of f at $x = 1$ and use it to estimate $f(1.1)$.
- Find $y = f(x)$ by solving the separable differential equation $\frac{dy}{dx} = \frac{3x^2 - 1}{2y}$ with the initial condition $f(1) = -4$.
- Use your solution from part c) to find $f(1.1)$.

FR2. 2019 Practice FR (Calculator)

The rate at which soft drink flows into the a storage container is modeled by the function R , where $R(t) = 120 + 25 \left(\cos\left(\frac{t}{2}\right) \right) \ln(t+10)$ gallons per minute, t is measured in minutes and $0 \leq t \leq 60$.

Bottles are filled using the beverage in the container at a rate modeled by $B(t) = 20\sqrt{100 - \frac{(t-30)^2}{10}}$ gallons per minute, for $0 \leq t \leq 60$. There are 2000 gallons of beverage in the container at time $t = 0$.

- Is the amount of beverage in the storage container increasing or decreasing at time $t = 10$ minutes? Give a reason for your answer.
- How many gallons of beverage flow into the storage container from time $0 \leq t \leq 30$?
- How many gallons of beverage are in the storage container at time $t = 40$ minutes?
- Find $B'(50)$ and, using correct units, explain the meaning of this in the context of this problem.

MC1. The length of the curve $y = \ln \sec x$ from $x = 0$ to $x = b$, where $0 < b < \frac{\pi}{2}$, may be expressed by which of the following integrals?

(A) $\int_0^b \sec x \, dx$

(B) $\int_0^b \sec^2 x \, dx$

(C) $\int_0^b (\sec x \tan x) \, dx$

(D) $\int_0^b \sqrt{1 + (\ln \sec x)^2} \, dx$

MC2. The area of the region enclosed by the graph of $y = x^2 + 1$ and the line $y = 5$ is

(A) $\frac{14}{3}$

(B) $\frac{16}{3}$

(C) $\frac{28}{3}$

(D) $\frac{32}{3}$

MC3. The base of a solid is the region enclosed by the graph of $y = e^{-x}$, the coordinate axes, and the line $x = 3$. If all plane cross sections perpendicular to the x -axis are squares, then its volume is

(A) $\frac{(1 - e^{-6})}{2}$

(B) $\frac{1}{2}e^{-6}$

(C) e^{-6}

(D) e^{-3}

MC4. CA A race car is traveling on a straight track at a velocity of 80 meters per second when the brakes are applied at time $t = 0$ seconds. From time $t = 0$ to the moment the race car stops, the acceleration of the race car is given by $a(t) = -6t^2 - t$ meters per second per second. During this time period, how far does the race car travel?

(A) 188.229 m

(B) 198.766 m

(C) 260.042 m

(D) 267.089 m